

A Camera Zoom-based Paper-Pencil Cipher Encryption Scheme atop Merkle–Hellman Knapsack Cryptosystem (Preliminary)

Gopal Anantharaman^{*1} and Prof. Jintai Ding²

¹KnotTheory.ai Inc., Fremont, CA, USA

²Xi'an Jiaotong–Liverpool University, Suzhou City, Jiangsu
Province, P. R. China

January 5, 2026

Abstract

A Symmetric Key Encryption scheme using Camera Zooming is presented using a familiar Paper-pencil cipher. The Camera can have a magnification/scaling up to some integer. The encrypting and decrypting systems are two hardware systems that are assumed to have the capability to zoom a given image with text from the resolution of a single character to a page by applying an appropriate scaling factor and an appropriate polynomial time zoom algorithm. Using the symmetric key, the Camera or a zooming algorithm implementation repeatedly zooms on different boxes in an image filled with cipher text and spurious redundant non-correlated pseudo-random data. Then, it decrypts the cipher using the Merkle-Hellman Knapsack Cryptosystem (MHKC) trapdoor algorithm or variants for efficient encryption/decryption. As shown, the MHKC algorithm's cryptanalysis vulnerability using Lattice-based LLL and other density attacks would not affect this scheme—as long as the key is private.

Keywords: Symmetric key encryption, Camera-based zooming, Merkle–hellman knapsack cryptosystem, Lattice-based cryptanalysis, gerrymandering geometry, steganography, optical air-gap communication

1 Introduction

Optical Character Recognition(OCR) is nearly reaching excellent accuracy in image and Handwriting recognition systems with the Advent of GenAI and

^{*}E-mail: gopal@knottheory.ai; Corresponding author; Contributed Paper @ SIGMAA BIG MathFest 2025

other advances in Computer Vision. Today, somebody can easily parse a hand-written note with OCR. Compelling evidence supports that Smart Camera-based hardware may soon take center stage in mainstream computing. Thus, camera algorithms, protocols, and security are in active research. In this paper, the familiar Paper-pencil Stencil encryption scheme is analyzed as a obfuscation layer over Merkle-Hellman Knapsack Crypto-system Trapdoor (MHKC) for a Camera reading the Ciphertext rather than a human. The MHKC is public key encryption scheme based on the subset sum problem that is in general NP-Hard. However, a special instance of the subset-sum, the super-increasing sequence, where each number is greater than the sum of all preceding numbers, is solvable and MHKC encryption scheme is based on a transformation of a super-increasing sequence problem to a knapsack trapdoor transformation that is secure. Knapsack based crypto-systems are highly relevant because the encryption is very fast - faster than RSA schemes. However, MHKC was broken by Shamir and Adleman because there exists a linearity of the modular transformation and the special structure in secret key. Even other variants using Chinese Remainder Theorem such as [7] are easily broken using Lattice reduction based cryptanalysis and LLL Algorithm. But, by partitioning the plain-text and splitting the partitions into geometric stencil pattern and adding combinatorial entropy before encryption effectively renders the LLL reduction impractical as the exact cipher-text is required for linear combination using MHKC public key for LLL algorithm initialization. Post LLL cryptanalysis, the resulting text will be a combination of pseudo-random letters along with the original plain-text. As we will analyze, Without the other secret key parameters such as the stencil locations or permutation number g , there is an extremely low probability (comparable to information theoretic limits) of an adversary gleaning the plain-text. The shared key includes the MHKC private key and is exchanged in a public medium using industry-standard key exchange algorithms such as ECDH or RSA.

2 Related Works

This work is not directly related to Visual Cryptography where the secret is obtained by combining portions of an image. This work is related to Steganography, in a way, where the resulting text, post decryption, conceals the plain-text. This work is not directly related to OCR and image analysis, however, post-decryption the handwriting used in the image could be used for authenticating the user if a neural-net is embedded in a camera to recognize the handwriting. This work is not directly related to [9] or other Optics based cryptographic techniques. However, uses optics as a medium of information transmission, i.e., for an optical air-gap communication without WiFi or other network connections.

3 Research Methodology

3.1 Paper-Pencil Cipher

A simple paper-pencil cipher is shown below with the hidden text in red blocks

Hello World

A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	L	L	R	R	B	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
Q	W	E	R	T	y	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	a	b	c	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	L	D	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	

With the Camera’s zoom capabilities, we could repeatedly zoom individually at boxes of varied lengths to figure out hidden text. The matrix also obfuscates the plain-text with random data. If the entire matrix had been encrypted, the cipher-text first had to be located, and then individual boxes had to be decrypted to get the plain-text. In this paper, we look at the encrypted text as an image/matrix in the Camera context, but nothing precludes the entire Matrix from being treated as a stream of bytes/bits. For example, By treating the above matrix as an unwound obfuscated message of length $rows \times columns$, the stencil sets, such as "RLD" above, become sets of indices into such message stream.

3.2 Definitions

Let Stencil Set S correspond to the exhaustive list of stencils of all shapes of the red boxes shown. For example, the text "RLD" is L-shaped.

We could also have other arbitrary shapes, such as disconnected ones containing the cipher-text.

Thus, Stencil set S is the set of all possible shapes to use. The stencil set S is

public.

Let M be the message. and $l = |M|$ denotes the length of a message.

Let R_l be a random partition of a random number $l_{max} \geq l$. i.e.

$$R_l = \{x_i \in Z^+ \mid \sum_{i=0}^n x_i = l_{max}\}$$

Let $S_{R_l} \subset S$ be a subset of shapes S such that there is a bijection $g : R_l \rightarrow S_{R_l}$. The requirement is that all the encrypted text of M fits within the Stencil sets in S_{R_l}

An example of an element of S_{R_l} for the picture above "RLD" Stencil set is

$$s_{rld} = \{(13, 12), (13, 13), (14, 13)\}$$

The first coordinate starts the Stencil assuming the fourth quadrant of the x-y plane with positive values for x, y . Here $s_{rld} \in S_{R_l}$. The encrypted text is split among the stencils and permuted, attaining enough permutation to be indistinguishable from surrounding random letters.

Let $\sigma(M)$ be the permutation of M post deciphering. we permute such that $\sigma^g(M) = M$.

The Secret key is the set $\{S_{R_l}, g, Private_{mhkc}, \sigma, g\}$ The underlying encryption algorithm is Public key Cryptosystem with keys $\{Public_{mhkc}, Private_{mhkc}\}$ of Merkle-Hellman Knapsack cryptosystem using MHKC trapdoor [6]

3.3 Encryption/Decryption

The encryption/decryption follows MHKC super-sequence-based knapsack cryptosystem as per [6] Or other variants such as [7] for the whole matrix with pre-filled permuted message $\sigma(M)$ in a chosen order at the appropriate stencil set. If a default order is not specified, then the start of the text and the stencil square with the start of the message are identified/included as part of the secret key. By default, the Camera is assumed to first zoom left-to-right top to bottom order for decryption. If a random zooming order on the stencils is enforced, for example, for the Matrix in the picture, stencil with W occurs before O, then there is an extra step of re-permuting the obtained text to glean the meaning of the full message in polynomial time. In such a scenario, the Secret key need not contain the permutation information.

4 Results and Discussion

4.1 Cryptanalysis

The Lenstra et al.'s LLL-lattice basis reduction [5] cryptanalysis for polynomial low-density attacks is still applicable (This is a 2-D Lattice), but security parameter IND-CPA (Indistinguishability, Chosen Plain text) is reconciled with

the random plain-text and the permutation of the actual plain-text in the deciphered matrix of cipher as shown in the above matrix.

4.2 Brief Strength Analysis

The strength of the encryption system lies in the randomness of the Stencil set S_{R_l} selected and the selection of a random partition R_l . And the indistinguishable structure that it offers. We focus on the probability of reaching the "actual" plain-text hidden within the matrix, given that the LLL-Lattice reduced-basis polynomial time reduction algorithm has been successfully run to decipher the Matrix (for MHKC or a Variant). This is a two-dimensional Lattice where even Gauss's basis reduction could be run in Polynomial Time to determine a reduced basis and, hence, a polynomial-time algorithm to arrive at the matrix.

The approximate probability of uniformly (uniform distribution) selecting a partition of a number $n = l_{max}$ is $P_1 = \frac{1}{P(n)}$ Where $P(n)$ is the classic mathematical partition function. Thus, approximately

$$P_1 = \frac{4n\sqrt{3}}{e^{\pi\sqrt{\frac{2n}{3}}}} \quad (1)$$

Next, the probability of selecting a stencil can be calculated as follows.

The number of elements of the partition so chosen, i.e., $k = |R_l| = |S_{R_l}|$, on average, is based on the probability distribution of the Message Space. The number of ways of selecting the stencil set is $\binom{|S|}{k}$; this is also the number of possible maps g . The probability of selecting a stencil set S_{R_l} is $P_2 = \frac{1}{\binom{|S|}{k}}$. Thus the overall probability of correctly getting the Stencil set containing the Plain text in the indistinguishable matrix is just

$$P = P_1 P_2 = \frac{4n\sqrt{3}}{e^{\pi\sqrt{\frac{2n}{3}}} \binom{|S|}{k}} \quad (2)$$

assuming statistically independent events. This means that by carefully choosing l_{max} and the stencil set size $|S|$ and with lower k , we can decrease the probability as low as possible, irrespective of the encryption scheme. In this case, the MHKC crypto-system is chosen to provide a fast encryption/decryption scheme. For Example, A message of length 100 has partition $P(100) = 190569292$ with a Stencil set S of 40000 and the selected stencil subset set size of S_{R_l} being 20 has $P = \frac{1}{(190569292 * \binom{40000}{20})} \approx 1.1666 \times 10^{-82}$ which is a very low probability and has $\approx \log_2 \frac{1}{P} = 82$ -bit security. In this case, we select 20 numbers x_i whose sum is 100.

4.3 Extensions and Complexity

In this scenario, the Matrix is assumed to be specific to a plain-text M . We can also have a predetermined matrix with the composition of various plain text,

where a set of plain texts share the same random matrix, or we can have a standard alphabetical/ASCII matrix. Similarly, the stencil sets can be the union of "disconnected" sets. The idea is to accommodate a part of the message corresponding to the value of x_i (as above) that makes up the selected partition in the same stencil set. Again. Nothing precludes not honoring that requirement. This is only for efficiency.

If there is no Matrix for each message, rather a standard ASCII Matrix is referred to by encrypting and decrypting systems, like an agreed reference Matrix containing all possible messages from message space of M , then the time complexity is the Complexity of the key exchange for Stencil set exchanges, as there is no encryption of the plain-text at all.

In all cases, The worst-case complexity is Polynomial in the size of the Matrix with no added random text. To randomly generate a partition of n , the time complexity is $O(n^{\frac{3}{4}})$ using Fristedt Algorithm [4].

To randomly generate a Stencil shape set, we could use Vitter's algorithm or variant [10] or [11] in $O(n)$ time.

Hence, a worst-case Polynomial asymptotic $O(n)$ is achievable, assuming MHKC with multiple iterations can be run at high speeds.

It is possible that the main Stencil sets can fully partition the Matrix into distinct parts. This means we bring in the geometry of gerrymandering to the stencils and their complexities. It is known that gerrymandering problems are at least NP-Hard, and there are reductions from a Rectilinear Minimum Steiner Tree NP-complete problem [8] to gerrymandering [1]. Here is an example with the actual message in Red and the main stencil sets that divide the Matrix in other colors. Here, we can see that additional stencils are required to locate the message within a main stencil set. This will further decrease the probability, as can be readily seen.

A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	L	L	R	R	B	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
Q	W	E	R	T	Y	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	A	B	c	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	L	D	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R	

Thus, Stencils provide enough obfuscation atop fast crypto-schemes.

5 Conclusion

5.1 Future Research directions

Security considerations for Higher-dimension for the stencil matrix is a possible scope for research with n -dimensional hyper-cube with unit character cells with higher dimensional stencil partitions. Gerrymandering geometry [3], topology and reduction to NP-Hard problems possibly offers more research directions towards converting the stencils and stencil geometry to purely stencil based public-key encryption scheme. For example a 9×9 grid has 706, 152, 947, 468, 301 possible equal sized stencils [2]. As an example, By carefully selecting a set of public and private stencil shapes that make up the 9×9 grid and publishing the grid and the public stencils as public key and to decrypt plain-text encoded in the public stencil shapes in the grid using the private key stencils' geometry offers scope for research for quantum-resistant geometric encryption schemes.

5.2 Practical limitations

5.2.1 OCR Latency and Error Correction

For a 100-byte message there is a latency of 10-30sec compared to QR codes' 100 – 500ms and other binary grids. In normal lighting, OCR accuracy drops

to $\approx 90\%$. To compensate, a standard approach would be to use Reed Solomon Forward Error Correction(RS-FEC) before scattering into stencils similar to QR codes. This adds $\approx 33\%$ overhead but greatly improves robustness to blur, glare and motion.

5.2.2 Camera Tilt and Stencils

Current stencil co-ordinates assume near perfect alignment, but real cameras introduce tilt, distortion and aberration leading to error reading neighboring cells. QR codes address this with corner markers and homographic transforms for perspective correction. Such anchor-based approach could make stencil access much more reliable using computer vision techniques.

5.2.3 AES, MHKC, Key sizes

For high-secure requirement MHKC could be replaced with AES-256 or stronger encryption schemes. For much more high-speed relaxed security environments, the encryption could be dropped altogether while retaining the stencil locations and permutation as the private key.

For Spatial stealth, a 100-byte message requires 250-byte key 2.5 : 1 ratio vs 0.32 : 1 ratio for AES-256. There might be more room to encode the partitions more efficiently. This is an implementation design choice.

5.3 Notes

Modern camera-based AI agent communication with vision helps process data asymmetrically using Cameras, Network protocols, and Human Collaboration. In such a Hybrid environment, seamless, secure transactions across domains are becoming a necessity rather than a theoretical interest while retaining existing Encryption schemes. The main takeaway being MHKC or other fast encryption schemes could be retrofitted with simple obfuscations to achieve reliable security mechanisms for Machine-to-Machine or Human-to-Machine communications.

6 Acknowledgments

Special thanks to Sarang Vehale (sarangvehale2@gmail.com) Grad Student at National Forensics Sciences University, Delhi, IN and Indian Institute of Technology, Chennai, IN for providing the Practical limitations details

References

- [1] Frunobulax (<https://mathoverflow.net/users/57621/frunobulax>). “Is this kind of Gerrymandering NP-complete?” In: (). URL: <https://mathoverflow.net/q/324524>.

- [2] Moon Duchin. “Geometry v GerryMandering”. In: *Scientific American* (Nov. 2018). URL: <https://www.scientificamerican.com/article/geometry-versus-gerrymandering/>.
- [3] Moon Duchin and Bridget Eileen Tenner. *Discrete geometry for electoral geography*. 2023. arXiv: 1808.05860 [physics.soc-ph]. URL: <https://arxiv.org/abs/1808.05860>.
- [4] Bert Fristedt. “The Structure of Random Partitions of Large Integers.” In: *Transactions of the American Mathematical Society* (1993), pp. 703–735. URL: <https://doi.org/10.2307/2154239>.
- [5] H. W. Jr; László Lovász Lenstra A. K; Lenstra. “Lenstra–Lenstra–Lovász lattice basis reduction algorithm”. In: *Math Ann* (1982). URL: <https://doi.org/10.1007/BF01457454>.
- [6] R. Merkle and M. Hellman. “Hiding information and signatures in trap-door knapsacks”. In: *IEEE Transactions on Information Theory* 24.5 (1978), pp. 525–530. DOI: 10.1109/TIT.1978.1055927.
- [7] Yasuyuki Murakami and Takeshi Nasako. “Knapsack Public-Key Cryptosystem Using Chinese Remainder Theorem”. In: *IACR Cryptol. ePrint Arch.* 2007 (2006), p. 107. URL: <https://api.semanticscholar.org/CorpusID:15121353>.
- [8] Rectilinear Steiner Tree. *Rectilinear Steiner Tree*. [Online; Accessed 2 Jul. 2024]. 2024. URL: http://en.wikipedia.org/wiki/Rectilinear_Steiner_tree.
- [9] Philippe Refregier and Bahram Javidi. “Optical image encryption based on input plane and Fourier plane random encoding”. In: *Opt. Lett.* 20.7 (Apr. 1995), pp. 767–769. DOI: 10.1364/OL.20.000767. URL: <https://opg.optica.org/ol/abstract.cfm?URI=ol-20-7-767>.
- [10] Jeffrey Scott Vitter. “An efficient algorithm for sequential random sampling”. In: *ACM Trans. Math. Softw.* 13.1 (Mar. 1987), pp. 58–67. ISSN: 0098-3500. DOI: 10.1145/23002.23003. URL: <https://doi.org/10.1145/23002.23003>.
- [11] Jeffrey Scott Vitter. “Faster methods for random sampling”. In: *Commun. ACM* 27.7 (July 1984), pp. 703–718. ISSN: 0001-0782. DOI: 10.1145/358105.893. URL: <https://doi.org/10.1145/358105.893>.